SmartCare

Online DoctorAppointmentBookingPlatform

Team Involvement & Contributions

Mayvel Technologies Applied Research Challenge — Management Round
Dhaanish Ahmed Institute of Technology, Coimbatore
March 2026

Name	Role	Contact
Sonaimuthu K	BD Lead & Research	sonaimuthu00700@gmail.com
Sandhiya G	References & Data Collection	sandhiyagunasekaran7@gmail.com
Syed Ahamed Jaafri A	Research Development Documentation	jaafri474@gmail.com
Aleef Nisha H	Backend Development	aleefnisha2@gmail.com
Karthika	Frontend Development	karthikaa3107@gmail.com

SONAIMUTHU K

B.Tech AI&DS; | Dhaanish AhmedInstituteofTechnology, Coimbatore

+91 6384625355 | sonaimuthu00700@gmail.com

BD LEAD & RESEARCH — Online Doctor Appointment Booking Platform

1. Problem Identification & Research

• Identified the core problem — patients facing long waiting times, manual phone-only booking, no doctor visibility, and lack of digital health records.

• Conducted real-world research on private clinics and hospitals in Virudhunagar district — how they currently manage appointments.

• Analyzed patient pain points and doctor challenges through observation and informal conversations.

2. Market Research & Target Client Analysis

• Researched India's digital health market — identified 27% CAGR growth and Rs. 485B market size by 2030.

• Identified target client segments: Private Hospitals, Clinics, Diagnostic Centres, and Telemedicine Startups.

• Independently compiled a dataset of 27 potential target clients across South India (Tamil Nadu, Kerala, Karnataka) — phone numbers, emails, websites, and BD status. Dataset: github.com/sonaimuthu-k/doctor-appointment-platform

• Benchmarked against competitors — Practo, mfine, Apollo 247 — to identify gaps our platform fills.

3. Business Model Design

• Designed 4-stream revenue model: SaaS Subscription, Commission per booking, Premium Doctor Listings, and Pharmacy Tie-ups.

• Proposed B2B-first approach — selling to clinics for faster revenue generation.

• Defined affordable pricing strategy for small clinics, scalable for hospitals.

4. BD Strategy

• Developed end-to-end BD strategy: LinkedIn outreach, cold calls, clinic visits with product demo, follow-up every 3 days.

• Rural BD strategy — WhatsApp outreach, Tamil language communication, and in-person visits in Virudhunagar and Rajapalayam.

• Referral strategy — leverage existing clients to bring new clinic partnerships.

5. Presentation & Documentation

• Prepared complete 6-slide PowerPoint presentation covering Problem, Features, App Flow, Business Model, and Conclusion.

• Coordinated team roles and documented project structure for GitHub repository.

• Led the pitch presentation to Mayvel Technologies management panel.

SANDHIYA G

Dhaanish AhmedInstituteofTechnology,Coimbatore

9360204944 | sandhiyagunasekaran7@gmail.com

REFERENCES & DATA COLLECTION — SmartCare Doctor Booking System

1. Data Collection Plan

- Conducted surveys with patients (age 18–65) to collect data on wait-time pain points and reminder preferences.
- Interviewed doctors to understand no-show rates and slot management challenges.
- Interviewed clinic admins on queue patterns and cancellation behavior.
- Conducted OPD floor observations to measure average wait time and queue behaviour.
- Collected UX feedback and SUS scores from beta prototype users.
- Gathered secondary benchmark data from WHO, NASSCOM, and healthcare IT reports.

2. Core Feature Research

- Researched Live Queue Tracking — token system, real-time updates via WebSocket (<1s latency), color-coded delay board (Green/Yellow/Red).
- Researched Auto Reminder System — SMS + WhatsApp 24 hrs before, 2-hr reminder with queue link, multilingual support (English, Tamil, Hindi).
- Researched Missed Appointment Recovery — no-show detection, instant next slot suggestion, waitlist auto-fill.

3. Tech Stack Research

- Documented technology recommendations: React Native/Flutter (Frontend), Node.js + Express (Backend), PostgreSQL + Redis (Database), Firebase/Socket.io (Realtime), Twilio + WhatsApp API (Notifications), Python + OpenAI API (AI Layer), AWS EC2 + S3 (Hosting).

4. References Compiled

- WHO (2023) — Global Health Observatory OPD Waiting Times.
- NASSCOM (2023) — Healthcare IT Adoption Report, India Digital Health.
- Practo Technologies (2023) — India Health Report, Digital Booking Trends.
- Twilio, Firebase, OpenAI, Meta/WhatsApp, TRAI — Technical API documentation.

SYED AHAMED JAAFRI A

B.Tech AI&DS | Dhaanish Ahmed Institute of Technology, Coimbatore

Ph: 6369407226 | jaafri474@gmail.com

Research Development Lead & Documentation — Online Doctor Booking System

1. Presentation Development & Technical Documentation

• Independently designed and developed the complete PowerPoint presentation for Mayvel Technologies management panel.

• Slide structure: Problem statement & pain points → Platform features → App flow & user journey → Business model → Conclusion & impact.

• Translated complex technical architecture and business logic into clear, visual communication.

2. System Design & Documentation

• **Designed Browse by Specialities logic — multi-parameter filter system (Experience, Gender, Fees, Language, Specialization).**

• Documented automated 24-hour reminder workflow — SMS Gateway integration flow diagram and appointment lifecycle documentation.

• Created separation of concerns architecture — frontend, backend, and service layer boundaries clearly defined.

3. AI & Automation Architecture

• Conceptualized AI Agent layer framework — **personalized reminder generation using Google Calendar API** integration.

• Designed dynamic **Email Template Generator** (live deployed project) and duplicate notification prevention logic.

• Planned extensible chatbot integration framework and SaaS expansion model documentation.

4. Documentation Deliverables

• App Workflow Document — end-to-end booking flow from patient login to confirmed appointment.

• Reminder System Design — time-based scheduler flow, SMS API integration, duplicate prevention guide.

• Feature Filter Design — multi-parameter doctor search guide for frontend and DB query implementation.

• AI Integration Blueprint — Google Calendar sync, personalized reminder logic, chatbot framework for future development.

5. Tools & Live Project

• Tools: n8n , PowerPoint, Figma/Draw.io, HTML+CSS, GitHub Pages, Google Calendar API, Technical writing.

• Live deployed project: Responsive Email Template Generator — syedjaafri.github.io/Responsive-Email-Template-Generator

ALEEF NISHA H

Reg No: 724022104003|Dhaanish AhmedInstituteofTechnology, Coimbatore

7994005052 | aleefnisha2@gmail.com

BACKEND DEVELOPER — Online Doctor Booking System

1. Core Backend Development (Java + Spring Boot)

• Built complete appointment processing engine — slot availability check, validation, user management using Java OOP principles.

• Developed RESTful APIs using Spring Boot (Controller → Service → Repository pattern) for seamless frontend-backend communication.

• Implemented Double-Booking Prevention Logic — checks existing appointments before confirming new slot using Java + MySQL transactions.

• Built Slot Validation Logic — validates doctor schedule and patient eligibility before booking confirmation.

2. Database Design (MySQL)

• Designed normalized relational schema with 5 core tables: users, doctors, specializations, slots, appointments.

• Implemented foreign key relationships ensuring data consistency and integrity across all entities.

• Wrote optimized SQL queries for slot availability checks and appointment management.

3. API Endpoints Developed

• POST /api/auth/register — Patient/Doctor registration with validation.

• POST /api/auth/login — JWT authentication and token generation.

• GET /api/doctors/{specialization} — Filter doctors by specialty.

• GET /api/slots/{doctorId}/{date} — Real-time slot availability.

• POST /api/appointments/book — Book with double-booking prevention.

• PUT /api/appointments/{id}/cancel — Cancel existing appointment.

KARTHIKA

Dhaanish Ahmed InstituteofTechnology, Coimbatore

6381888149 | karthikaa3107@gmail.com

FRONTEND DEVELOPER — Online Doctor Booking System

1. Core Frontend Development (HTML + CSS + JavaScript)

- Built complete responsive frontend using HTML5, CSS3, and JavaScript — mobile-first design ensuring seamless experience across all devices.
- Implemented real-time form validation, dynamic DOM manipulation, and event-driven interactions for smooth booking flow.
- Integrated Fetch API calls to Spring Boot REST endpoints for dynamic data loading.
- Implemented JWT token storage and auth headers for secure frontend-backend communication.

2. Key Screens Developed

- Home / Landing Page — Hero section, navigation bar, call-to-action buttons, speciality cards.
- Patient Registration — Real-time field validation, error messages, submit handler.
- Doctor Listing Page — Filter by speciality, search bar, doctor cards with ratings and fees.
- Appointment Booking Page — Date picker, available slot display, booking confirmation modal.
- Patient Dashboard — Upcoming and past appointments, cancel option, profile section.
- Doctor Profile Page — Bio, experience, availability schedule, book button.

3. UI/UX Design Highlights

- Mobile-first responsive layout with consistent color palette and font system.
- Card-based doctor listing for easy scanning, clear CTA buttons at every step.
- Loading indicators during API calls, empty state messages, confirmation modals before booking.
- Cross-browser compatibility and accessibility best practices throughout.